

Company/Business System Network Design

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MINI LAB PROJECT REPORT

This Report Presented in Partial Fulfillment of the course **CSE313: Computer Networks Lab** in the Computer Science and Engineering Department



DAFFODIL INTERNATIONAL UNIVERSITY
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DECLARATION

We hereby declare that this lab project has been done by us under the supervision of **Ms. Syada Tasmia Alvi, Lecturer**, Department of Computer Science and Engineering, Daffodil International University. We also declare that neither this project nor any part of this project has been submitted elsewhere as lab projects.

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COURSE & PROGRAM OUTCOME

The following course have course outcomes as following:

Table 1: Course Outcome Statements

CO's	Statements
CO1	Understand the basic knowledge of networking fundamentals, economic factors, and simulation tools for modern communication systems.
CO2	Analyze an adaptable approach to network configuration and optimization by analyzing IP address allocation, evaluating current and emerging communication protocols to configure routers, switches, and servers.
CO3	Design diverse network topologies and routing protocols using Packet Tracer through collaborative projects and presentations, effectively communicate technical decisions and justifications regarding network design and optimization.

Table 2: Mapping of CO, PO, Blooms, KP and CEP

CO	PO	Blooms	KP	CEP
CO1	PO5	C2	KP4	EP1
CO2	PO2	C3, P3	KP3	EP2
CO3	PO3	P2, A2	KP8	EP4

The mapping justification of this table is provided in section 4.3.1, 4.3.2 and 4.3.3.

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Chapter 1

Introduction

1.1 Introduction

In the modern digital landscape, where businesses depend on efficient communication and secure data management, a well-designed and reliable network is vital. The **Secure Company Network System (SCNS)** offers a comprehensive solution to ensure the safety, performance, and reliability of a company's network infrastructure. SCNS is built to protect critical data, facilitate smooth communication, and prevent unauthorized access through advanced security measures such as encryption, access controls, and real-time monitoring.

The SCNS integrates cutting-edge technology to provide a secure environment that detects and addresses threats proactively. By isolating sensitive traffic, segmenting departments, and prioritizing critical services, it creates an optimized and scalable network. With real-time monitoring tools, SCNS identifies potential problems instantly, ensuring quick resolution to maintain operational continuity.

The primary goal of SCNS is to create a secure, efficient, and future-ready network that not only shields the organization from cyber threats but also supports its growth and enhances productivity. This system enables organizations to work seamlessly while maintaining high levels of security and reliability, providing a strong foundation for success in a competitive business environment.

1.2 Motivation

In today's digital landscape, companies are increasingly reliant on networked systems for operations, collaboration, and data management. This project is motivated by:

1. The increasing threats to network security due to cyberattacks.
2. The necessity of uninterrupted communication and data flow for operational efficiency.
3. The strategic importance of aligning network capabilities with the company's growth objectives.

By leveraging Cisco Packet Tracer, this project aims to simulate and develop a solution that addresses these challenges effectively.

1.3 Objectives

The main objectives of the Secure Company Network System (SCNS) project are designed to meet

the specific needs of the company's growing network infrastructure. The project focuses on:

- **Hierarchical Design:** Implement a layered network with redundancy for reliability.
- **ISP Connectivity:** Ensure internet reliability by connecting to at least two ISPs.
- **Wireless Networks:** Deploy department-specific wireless networks.
- **VLANs and Subnets:** Allocate distinct VLANs and subnets for secure communication.
- **Advanced Protocols:** Configure Secure Shell (SSH) for secure access and Port Address Translation(PAT) for efficient IP use.
- **Scalability:** Build a network that supports current needs and future growth.

1.4 Feasibility Study

Technical

Feasibility:

The use of Cisco Packet Tracer allows for detailed network simulations, making the design process technically viable. Advanced tools like VLANs, and routing protocols will ensure robust performance.

Economic

Feasibility:

By simulating the network design before physical implementation, costs are minimized, and potential missteps are avoided.

Operational

Feasibility:

The proposed design is user-friendly and ensures minimal disruption to daily operations during and after implementation.

Legal

Feasibility:

The project adheres to industry-standard network protocols and complies with relevant cybersecurity regulations.

1.5 Gap Analysis

Current State:

Security: Existing security measures (firewalls, ACLs) are outdated and lack strong encryption protocols, leaving sensitive data exposed.

Performance: Bandwidth is limited in key areas, causing congestion, and there is no QoS for prioritizing critical applications like VoIP.

Reliability: Lack of redundancy in core and distribution layers creates single points of failure, risking network downtime.

Desired State:

Security: Implement strong encryption, improve VLAN segmentation, and deploy advanced IDS/IPS for better threat detection.

Performance: Upgrade critical bandwidth links, implement QoS to prioritize vital applications, and monitor network performance in real-time.

Reliability: Add redundancy in core/distribution layers, implement failover mechanisms, and ensure backup power for key devices.

Gaps

- **Security:** No end-to-end encryption, insufficient access control, and lack of advanced threat detection.
- **Performance:** Insufficient bandwidth and QoS policies.
- **Reliability:** Single points of failure and no real-time monitoring.

Action Plan

- **Security:** Upgrade encryption and implement advanced security tools like IPS.
- **Performance:** Increase bandwidth and implement QoS.
- **Reliability:** Add redundancy and failover mechanisms.

1.6 Project Outcome

The Secure Company Network System (SCNS) project will deliver a robust and secure network design that meets the immediate and future needs of the company. Key outcomes include:

1. A fully simulated and optimized network design using Cisco Packet Tracer.
2. Enhanced network security through the implementation of VLANs, Advanced Protocols.
3. Seamless communication and data transfer supported by efficient routing and switching protocols.
4. A scalable network infrastructure capable of adapting to business growth.
5. Documentation and simulation results for future reference and implementation guidance.

This project positions the organization to operate efficiently and securely in its new facility, ensuring long-term resilience and adaptability.

Chapter 2

Proposed Methodology/Architecture

2.1 Requirement Analysis & Design Specification

2.1.1 Overview

The "Company System Network Design" in Cisco Packet Tracer follows a hierarchical model with three layers for efficiency, scalability, and redundancy.

1. **Core Layer:** The Core Layer acts as the backbone of the network, ensuring high-speed, reliable data routing across the entire infrastructure. It includes two routers and multilayer switches that handle both routing and switching, enabling seamless communication between different network segments. With its focus on performance and uptime, this layer offers redundancy and high availability, minimizing the risk of network downtime and ensuring uninterrupted data flow.
2. **Distribution Layer:** The Distribution Layer connects various departments, manages traffic between different VLANs, and ensures secure and efficient communication. It features switches that link departmental VLANs and enables inter-VLAN routing, allowing different segments of the network to communicate with one another. This layer also applies Access Control Lists (ACLs) for enhanced security and utilizes Quality of Service (QoS) to prioritize critical applications, ensuring optimal performance for time-sensitive services like VoIP.
3. **Access Layer:** The Access Layer connects end-user devices, such as PCs, wireless access points, and printers, to the network, providing the interface for user access. Switches and access points in this layer facilitate seamless device connectivity and enable efficient data transfer to higher layers. Each device is assigned to a specific VLAN for better organization and security, while port security ensures that only authorized devices can connect to the network, preventing unauthorized access and enhancing network integrity.

2.1.2 Proposed Methodology/ System Design

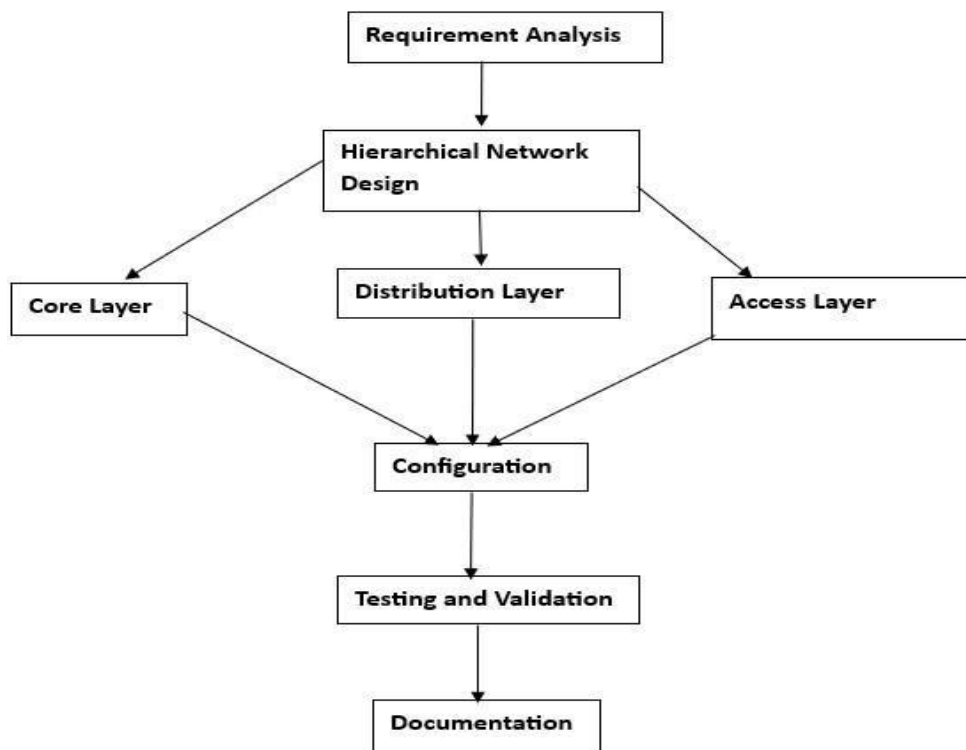
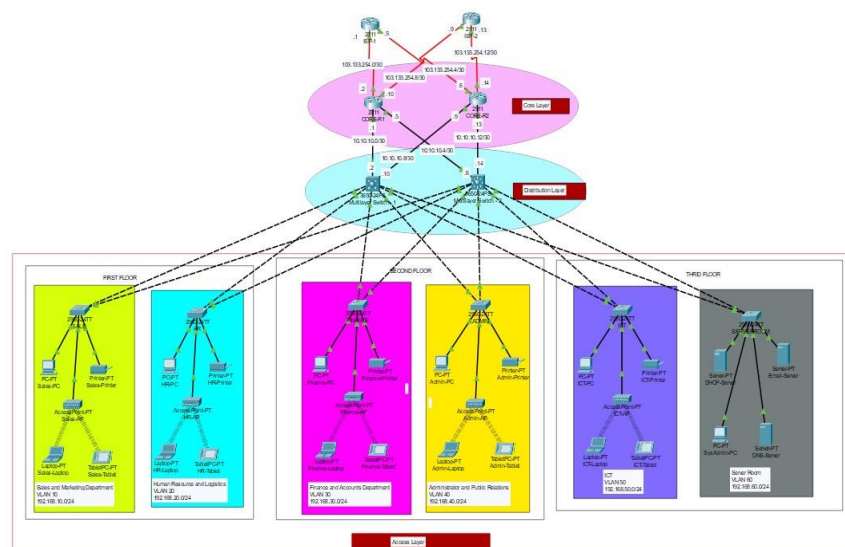


Figure 2.1: This is a sample diagram

2.1.3 UI Design Diagram:



Page Layout

- Use a **two-column structure**:
 - **Left Column**: Explanatory text and legends.

- **Right Column:** The network diagram for visual reference.
- Add section headers (e.g., Overview, Components, VLAN Structure, Recommendations).

Color Palette

- Use a **professional and muted palette** with soft gradients to maintain readability in print and digital formats.
- Highlight key sections (Core, Distribution, Access layers) using subtle colors (e.g., light gray, soft blue, or pastel shades).

Diagram Integration

- Embed the network diagram prominently in the middle of the report or as a centerpiece.
- Annotate the diagram with callouts for VLANs, devices, and layers.
- Include a **legend** beneath the diagram for a quick reference to symbols and colors.

Typography

- Use modern, legible fonts:
 - Headers: Sans-serif, bold (e.g., Arial or Calibri).
 - Body text: Clean serif fonts (e.g., Times New Roman) for readability.
- Use consistent font sizes for headers and subheaders.

Visual Enhancements

- Incorporate icons for device types (e.g., router, switch, PC, printer).
- Show structured connections with clean, straight lines or curves between layers.

Additional Features

- Include a **table of contents** for report navigation.
- Add **dynamic elements** like numbered sections or clickable links for digital formats.
- Provide space for a summary and conclusions at the end.

2.2 Overall Project Plan

Project Initialization

- Define the project scope and objectives.
- Identify key stakeholders and form the project team.

Requirement Analysis

- Gather network requirements, including scalability, security, and departmental needs.
- Analyze the existing network infrastructure (if applicable).

Design Phase

- Create a hierarchical network design with core, distribution, and access layers.
- Plan IP addressing, VLANs, routing protocols, and security measures.
- Ensure redundancy and scalability in the design.

Simulation and Configuration

- Use Cisco Packet Tracer to simulate the network architecture.
- Configure network devices, including routers, switches, and access points.
- Set up security protocols such as ACLs, SSH, and PAT.

Testing Phase

- Test security measures to prevent unauthorized access.
- Verify connectivity between VLANs and proper routing configurations.

Evaluation and Optimization

- Analyze test results to identify bottlenecks or vulnerabilities.
- Optimize configurations for better performance and security.

Implementation Planning

- Develop a step-by-step implementation guide for deploying the network in the real environment.
- Provide training for IT staff on managing and maintaining the network.

Documentation

- Prepare detailed project documentation, including design diagrams, configurations, and testing results.

Chapter 3

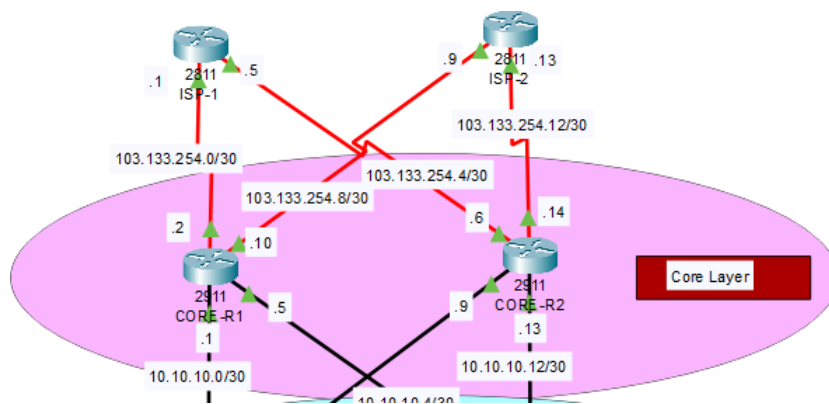
Implementation and Results

3.1 Implementation

1. Routers (4):

- 2 ISP router for upstream connectivity.
- Positioned at the core layer for redundancy.
- Connect to both ISPs for internet connectivity.
- Configured with static, public IP addresses from ISPs.

Figure:



Code: For router 1,

```

=====
Core Router - 1
=====
Basic Setting
-----
en
config

hostname CORE-R1
line console 0
password cisco
login
exit

enable password cisco
no ip domain-lookup
banner motd #No Unauthorised Access!!!#
service password-encryption

do wr

SSH Configuration
-----
ip domain name cisco.net
username admin password cisco
crypto key generate rsa
1024
line vty 0 15
login local
transport input ssh
exit

ip ssh version 2

do wr

Assigning IP address
-----
int giga0/0
ip address 10.10.10.1 255.255.255.252
no shutdown
exit
do wr

int giga0/1
ip address 10.10.10.5 255.255.255.252
no shutdown
exit
do wr

int se0/2/0
ip address 103.133.254.1 255.255.255.252
no shutdown
clock rate 64000
exit
do wr

int se0/2/1
ip address 103.133.254.10 255.255.255.252
no shutdown
clock rate 64000
exit
do wr

OSPF Configuration ( OSPF 10 is the process ID )
-----
router ospf 10
router-id 3.3.3.3
network 10.10.10.0 0.0.0.3 area 0
network 10.10.10.4 0.0.0.3 area 0
network 103.133.254.0 0.0.0.3 area 0
network 103.133.254.8 0.0.0.3 area 0

do wr
exit

```

```

PAT Configuration
-----
ip nat inside source list 1 int se0/2/0 overload
ip nat inside source list 1 int se0/2/1 overload

access-list 1 permit 192.168.10.0 0.0.0.255
access-list 1 permit 192.168.20.0 0.0.0.255
access-list 1 permit 192.168.30.0 0.0.0.255
access-list 1 permit 192.168.40.0 0.0.0.255
access-list 1 permit 192.168.50.0 0.0.0.255
access-list 1 permit 192.168.60.0 0.0.0.255

int range gig0/0-1
ip nat inside
exit

int se0/2/0
ip nat outside
int se0/2/1
ip nat outside
exit

do wr

Default static route
-----
ip route 0.0.0.0 0.0.0.0 se0/2/0 // If problem use: ip route 0.0.0.0 0.0.0.0 se0/2/0 70
ip route 0.0.0.0 0.0.0.0 se0/2/1 70

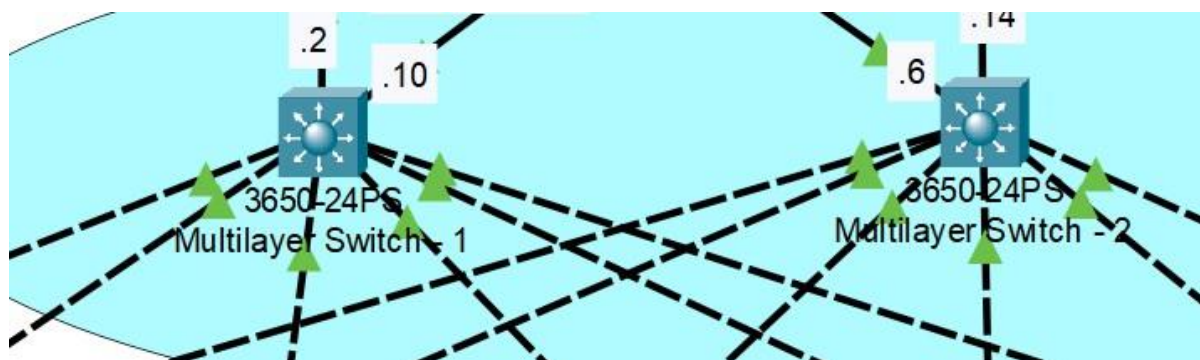
do wr

```

2. Multilayer Switches (2):

- Deployed at the core layer to provide redundancy and efficient routing.
- Configured for both switching and routing functionalities.
- Assigned IP addresses to enable inter-VLAN routing.

Figure:



Code: For 1 switch

```

=====
Multilayer Switch - 1
=====
Basic Setting
-----
en
config

hostname Mlt-SW1
line console 0
password cisco
login
exit

enable password cisco
no ip domain-lookup
banner motd #No Unauthorised Acces!!!#
service password-encryption

do wr

SSH Configuration
-----
ip domain name cisco.net
username admin password cisco
crypto key generate rsa
1024
line vty 0 15
login local
transport input ssh
exit

ip ssh version 2

do wr

gig1/0/3 to gig1/0/8 is trunk port and connect to VLAN
-----
int range gig1/0/3-8

do wr

int gig1/0/2
ip address 10.10.10.10 255.255.255.252
no shutdown
exit
do wr

OSPF Configuration (ospf 10 is process ID)
-----
ip routing
router ospf 10
router-id 2.2.2.2
network 192.168.10.0 0.0.0.255 area 0
network 192.168.20.0 0.0.0.255 area 0
network 192.168.30.0 0.0.0.255 area 0
network 192.168.40.0 0.0.0.255 area 0
network 192.168.50.0 0.0.0.255 area 0
network 192.168.60.0 0.0.0.255 area 0
network 10.10.10.0 0.0.0.3 area 0
network 10.10.10.8 0.0.0.3 area 0

do wr

Inter-VLAN configuration
-----
int vlan 10
no shutdown
ip address 192.168.10.1 255.255.255.0
ip helper-address 192.168.60.2
exit

int vlan 20
no shutdown
ip address 192.168.20.1 255.255.255.0
ip helper-address 192.168.60.2
exit

int vlan 30
no shutdown
ip address 192.168.30.1 255.255.255.0

int range gig1/0/3-8
switchport mode trunk

vlan 10
name Sales
vlan 20
name HR
vlan 30
name Finance
vlan 40
name Admin
vlan 50
name ICT
vlan 60
name ServerRoom

exit

do wr

Making gig1/0/1-2 into a layer 3 interface
-----
int range gig1/0/1-2
no switchport

exit
do wr

Assigning IP address
-----
int gig1/0/1
ip address 10.10.10.2 255.255.255.252
no shutdown
exit
do wr

int gig1/0/2
ip address 10.10.10.10 255.255.255.252
no shutdown
int vlan 20
no shutdown
ip address 192.168.20.1 255.255.255.0
ip helper-address 192.168.60.2
exit

int vlan 30
no shutdown
ip address 192.168.30.1 255.255.255.0
ip helper-address 192.168.60.2
exit

int vlan 40
no shutdown
ip address 192.168.40.1 255.255.255.0
ip helper-address 192.168.60.2
exit

int vlan 50
no shutdown
ip address 192.168.50.1 255.255.255.0
ip helper-address 192.168.60.2
exit

int vlan 60
no shutdown
ip address 192.168.60.1 255.255.255.0
exit

do wr

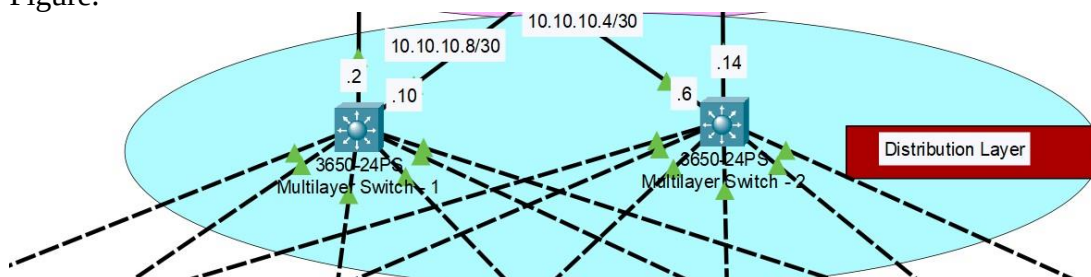
Default static route
-----
ip route 0.0.0.0 0.0.0.0 gig1/0/1

```

3. Distribution Layer Switches (Multiple):

- Connect individual departments to the core layer.
- Facilitate communication within respective VLANs.

Figure:

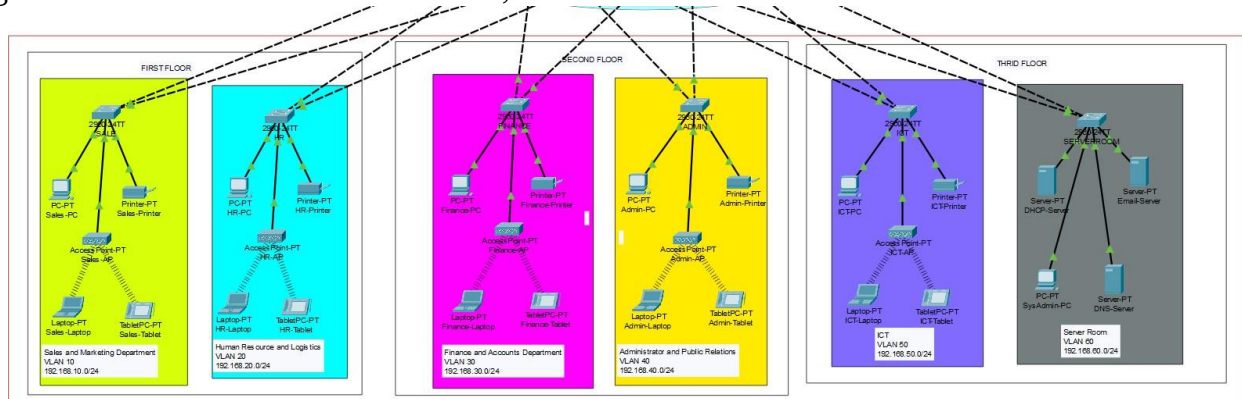


4. End-User Devices (PCs):

- Deployed at the access layer.

- Connected to distribution layer switches for departmental access.

Figure: For first and second floor Devices,



Code:

Sales & Marketing

```
=====
Sales and Marketing
=====
Basic Setting
-----
do
config t
hostname Sales-SW
banner motd #No Unauthorised Acces!!!#
no ip domain-lookup
line console 0
passw cisco
login
exit

enable password cisco
service password-encryption
exit
wr * write memory

VLAN and Trunk port and Access port
-----
int range fa0/1-2
switchport mode trunk
exit

vlan 10
name Sales
exit

int range fa0/3-24
switchport mode access
switchport access vlan 10
exit

do wr

/* Putting other giabitEthernet port to unused vlan with shutdown */
vlan 99
name Blackhole
exit

int range gig0/1-2
switchport mode access
switchport access vlan 99
shutdown
exit

do wr
```

HR & Logistic

```
=====
HR and Logistic
=====
Basic Setting
-----
do
config
hostname HR-SW
line console 0
password cisco
login
exit

enable password cisco
no ip domain-lookup
banner motd #No Unauthorised Acces!!!#
service password-encryption

do wr

VLAN and Trunk port and Access port
-----
int range fa0/1-2
switchport mode trunk
exit

vlan 20
name HR
vlan 99
name Blackhole
exit

int range fa0/3-24
switchport mode access
switchport access vlan 20
exit

int range gig0/1-2
switchport mode access
switchport access vlan 99
exit

do wr
```

Finance & accounts

Administrator and Public

Finance and Accounts	Administrator and Public
<pre> Basic Setting ----- en config hostname Finance-SW line console 0 password cisco login exit enable password cisco no ip domain-lookup banner motd #No Unauthorised Acces!!!# service password-encryption do wr VLAN and Trunk port and Access port ----- int range fa0/1-2 switchport mode trunk exit vlan 30 name Finance vlan 99 name BlackHole exit int range fa0/3-24 switchport mode access switchport access vlan 30 exit int range gig0/1-2 switchport mode access switchport access vlan 99 shutdown exit do wr Switchport security to finance department ----- int range fa0/3-24 Switchport port-security switchport port-security maximum 1 (// Means 1 user at a time) switchport port-security mac-address sticky switchport port-security violation shutdown exit do wr Excluding int fa0/6 from port-security ----- int fa0/6 no Switchport port-security no switchport port-security maximum 1 no switchport port-security mac-address sticky no switchport port-security violation shutdown exit </pre>	<pre> Basic Setting ----- en config hostname Admin-SW line console 0 password cisco login exit enable password cisco no ip domain-lookup banner motd #No Unauthorised Acces!!!# service password-encryption do wr VLAN and Trunk port and Access port ----- int range fa0/1-2 switchport mode trunk exit vlan 40 name Admin vlan 99 name BlackHole exit int range fa0/3-24 switchport mode access switchport access vlan 40 exit int range gig0/1-2 switchport mode access switchport access vlan 99 shutdown exit do wr </pre>

ICT

ICT
<pre> Basic Setting ----- en config hostname ICT-SW line console 0 password cisco login exit enable password cisco no ip domain-lookup banner motd #No Unauthorised Acces!!!# service password-encryption do wr VLAN and Trunk port and Access port ----- int range fa0/1-2 switchport mode trunk exit vlan 50 name ICT vlan 99 name BlackHole exit int range fa0/3-24 switchport mode access switchport access vlan 50 exit int range gig0/1-2 switchport mode access switchport access vlan 99 shutdown exit do wr </pre>

5. Cisco Access Points (APs):

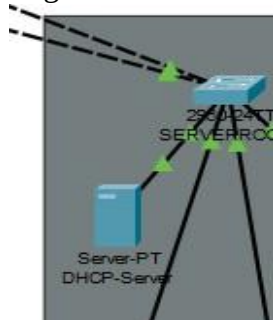
- Positioned at the access layer to provide wireless connectivity.

- Ensure wireless network availability in each department.

6. DHCP Servers (1):

- Located in the server room.
- Dynamically allocate IP addresses to end-user devices.

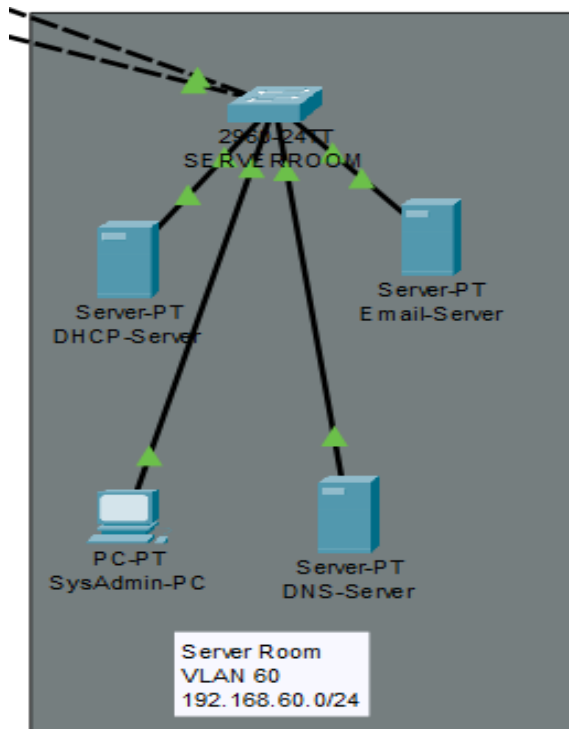
Figure:



7. Server Room Devices (Servers, etc.):

- DNS server, HTTP server etc.
- Devices in the server room are allocated static IP addresses.
- These devices may include servers, storage units, and networking equipment.

Figure & code:



```

=====
Server Room
=====
Basic Setting
-----
en
config

hostname ServerRoom-SW
line console 0
password cisco
login
exit

enable password cisco
no ip domain-lookup
banner motd #No Unauthorised Acces!!!#
service password-encryption

do wC

VLAN and Trunk port and Access port
-----
int range fa0/1-2
switchport mode trunk
exit

vlan 60
name ServerRoom
vlan 99
name BlackHole
exit

int range fa0/3-24
switchport mode access
switchport access vlan 60
exit

int range gig0/1-2
switchport mode access
switchport access vlan 99
shutdown
exit

do wC

```

3.2 Performance Analysis

1. Scalability

- **Network Design:**
 - The hierarchical structure allows for scalability, as additional devices can be integrated into the access layer without major changes to the core and distribution layers.
 - VLAN segmentation helps manage traffic growth efficiently by isolating broadcast domains.
- **Recommendations:**
 - Ensure core and distribution devices have sufficient capacity to handle increased traffic.

2. Bandwidth Utilization

- **Current Setup:**
 - Evaluate the bandwidth capacity of the links between core, distribution, and access layers.
 - Consider if uplinks (e.g., between core routers and distribution switches) are running at 1 Gbps or 10 Gbps.
- **Potential Issues:**
 - Bottlenecks may occur if uplink speeds are insufficient for high traffic volumes.
- **Recommendations:**
 - Upgrade critical links to higher speeds if necessary (e.g., 10 Gbps for core to distribution).

3. Redundancy and Reliability

- **Design Strengths:**
 - The presence of multiple core devices indicates redundancy, reducing single points of failure.
- **Potential Risks:**
 - Check if redundancy is implemented at the distribution and access layers.
- **Recommendations:**
 - Introduce backup links or devices at distribution and access layers for enhanced reliability.

4. Latency

- **Factors Affecting Latency:**
 - The hierarchical design minimizes latency by keeping traffic localized within VLANs.
 - Cross-VLAN traffic handled by inter-VLAN routing can introduce delays.
- **Recommendations:**
 - Ensure efficient routing protocols (e.g., OSPF, EIGRP) are used for inter-VLAN communication.
 - Use Quality of Service (QoS) for prioritizing time-sensitive traffic like VoIP.

5. Security

- **Current Measures:**
 - VLANs inherently enhance security by segregating traffic.
 - Distribution layer likely applies Access Control Lists (ACLs) for traffic filtering.
- **Recommendations:**
 - Implement firewalls and intrusion prevention systems (IPS) at key network points.
 - Use network monitoring tools to detect and mitigate security threats.

6. Device Performance

- **Core Devices:**
 - Core routers must handle high throughput and multiple routing table entries without performance degradation.

- **Access Devices:**
 - Ensure switches and access points can support connected devices without dropping packets.
- **Recommendations:**
 - Regularly monitor CPU and memory utilization on all devices.
 - Upgrade older devices if resource utilization is consistently high.

7. Traffic Analysis

- **Tools:**
 - Use traffic monitoring tools (e.g., Cisco NetFlow, Wireshark) to evaluate network usage.
- **Key Metrics:**
 - Throughput: Measure actual data transfer rates.
 - Packet Loss: Assess if packet drops are occurring due to congestion or faulty links.
- **Recommendations:**
 - Optimize traffic flow by redistributing VLANs or adjusting routing paths if bottlenecks are detected.

3.3 Results and Discussion

Result:

1. **Data Transmission:**
 - Data is transmitted across the network using a hierarchical model:
 - **Within the Same VLAN:** Data flows directly between devices in the same VLAN, minimizing latency as there is no need for inter-VLAN routing. For instance, HR PCs communicate directly with the HR printer within the same broadcast domain.
 - **Inter-VLAN Communication:** Devices in different VLANs communicate via Layer 3 switches or routers at the distribution layer. The router or Layer 3 switch processes the traffic by routing it between VLANs based on predefined Access Control Lists (ACLs).
 - **End-to-End Path:**
 - Data originates from an end device at the access layer (e.g., a PC or printer).
 - It passes through the access switch, forwarded to the distribution switch, and routed to its destination.
 - For external data (e.g., internet access), traffic flows through the core layer, where routing to the external gateway occurs.
2. **Network Performance:**
 - **Intra-VLAN Traffic:** Smooth transmission with low latency and minimal packet loss due to localized traffic handling within VLANs.
 - **Inter-VLAN Traffic:** Effective communication with slight delays due to routing at the distribution layer.
 - **External Traffic:** Internet-bound data is efficiently routed through the core layer, leveraging high-speed links to the external gateway.
3. **Bandwidth and Resource Utilization:**
 - Traffic is efficiently distributed across the network, with high-speed links at the core layer preventing bottlenecks.
 - Access and distribution switches handle device loads effectively, with sufficient capacity for current traffic demands.
4. **Security:**
 - VLAN segmentation ensures isolated traffic, reducing the risk of broadcast storms and

- unauthorized access between departments.
- ACLs at the distribution layer regulate inter-VLAN and external communication, enhancing security.

Screenshot:

For same VLAN:

Realtime Simulation											
Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete	
	Successful	Sales...	Sales-Printer	ICMP		0.000	N	0	(edit)	(delete)	
	Successful	HR-PC	HR-Printer	ICMP		0.000	N	1	(edit)	(delete)	
	Successful	Finan...	Finance-P...	ICMP		0.000	N	2	(edit)	(delete)	
	Successful	HR-PC	HR-Printer	ICMP		0.000	N	3	(edit)	(delete)	

For Different VLAN:

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete	
	Failed	Sales...	HR-Printer	ICMP		0.000	N	0	(edit)	(delete)	
	Failed	Finan...	Admin-PC	ICMP		0.000	N	1	(edit)	(delete)	
	Failed	ICT-PC	Admin-Pri...	ICMP		0.000	N	2	(edit)	(delete)	

Discussion

- **Data Flow Optimization:**
 - The hierarchical design ensures efficient data transmission by localizing traffic where possible (within VLANs) and routing it through optimized paths for inter-VLAN or external traffic.
 - Introducing Quality of Service (QoS) will prioritize time-sensitive data, such as VoIP and video conferencing, improving transmission quality.
- **Redundancy and Reliability:**
 - Core layer redundancy ensures uninterrupted data flow, even in the event of link or device failures.
 - Adding redundant links at the access and distribution layers would further enhance data transmission reliability.
- **Security Considerations:**
 - Firewalls and intrusion prevention systems can secure data transmission against external threats.
 - Regular updates to ACLs and VLAN configurations will maintain secure and efficient data flows.

Chapter 4

Engineering Standards and Mapping

4.1 Impact on Society, Environment and Sustainability

1. **Enhanced Communication:** Reliable networks improve communication within and outside the organization.
2. **Increased Productivity:** Streamlined operations lead to higher workplace efficiency.
3. **Reduced Carbon Footprint:** Implementing energy-efficient devices minimizes environmental impact.
4. **Job Opportunities:** Demand for network management and security creates employment.

4.1.1 Impact on Life

1. **Improved Connectivity:** Enables seamless access to information and resources.
2. **Better Work Environment:** Secure and efficient systems reduce downtime and stress.
3. **Data Security:** Protects sensitive personal and organizational data from breaches.

4.1.2 Impact on Society & Environment

1. **Positive Social Impact:** Enhances organizational performance, benefiting clients and stakeholders.
2. **Environmental Considerations:** Promotes eco-friendly practices through efficient resource management.
3. **Social Equity:** Ensures fair access to technology for all departments and teams.

4.1.3 Ethical Aspects

1. **Data Privacy:** Adheres to ethical standards in securing personal and organizational information.
2. **Equal Access:** Ensures fair and unbiased access to network resources.
3. **Transparency:** Maintains accountability in system design and implementation.

4.1.4 Sustainability Plan

1. **Energy Efficiency:** Use low-power devices and optimize network configurations.
2. **Scalability:** Design the network to adapt to future growth with minimal redesign.
3. **Regular Maintenance:** Schedule updates and audits to ensure long-term performance.
4. **Training:** Provide staff training to promote sustainable and ethical use of the system.
5. **Reuse of Resources:** Repurpose existing hardware where feasible to reduce waste.

4.2 Project Management and Teamwork

1. Jahangir Alam Jibon → Implementation, Results and Complex Engineering, Proposed Methodology, Introduction Part of Lab Report and Conclusion Part of Lab Report.

4.3 Complex Engineering Problem

4.3.1 Mapping of Program Outcome

In this section, provide a mapping of the problem and provide a solution with targeted Program Outcomes (PO's).

Table 4.1: Justification of Program Outcomes

COs	COs	POs	Learning domain	Knowledge Profile	Complex Engineering Problem	Complex Engineering Activities
CO1	Understand the basic knowledge of networking fundamentals, economic factors, and simulation tools for modern communication system.	PO5	C2	K4	EP1	EA1
CO2	Analyze an adaptable approach to network configuration and optimization by analyzing IP address allocation, evaluating current and emerging communication protocols to configure routers, switches, and servers.	PO2	C3,P3	K3	EP2	EA2
CO3	Design diverse network topologies and routing protocols using Packet Tracer through collaborative projects and presentations, effectively communicate technical decisions and justifications regarding network design and optimization.	PO3	A2,P2	K8	EP4	EA4

4.3.2 Complex Problem Solving

In this section, provide a mapping with problem solving categories. For each mapping add subsections to put rationale (Use Table 4.2). For P1, you need to put another mapping with

4.3. Complex Engineering Problem Knowledge profile and rational thereof.

Table 4.2: Mapping with complex problem solving.

EP1 Depth of Knowledge	EP2 Range of Conflicting Require- ments	EP3 Depth of Analysis	EP4 Familiarity of Issues	EP5 Extent of Applicable Codes	EP6 Extent of Stake- holder Involve- ment	EP7 Inter- dependence
✓	✓		✓			

4.3.3 Engineering Activities

In this section, provide a mapping with engineering activities. For each mapping add subsections to put rationale (Use Table 4.3).

Table 4.3: Mapping with complex engineering activities.

EA1 Range of re- sources	EA2 Level of Interac- tion	EA3 Innovation	EA4 Consequences for society and envi- ronment	EA5 Familiarity
✓	✓		✓	

Chapter 5

Conclusion

5.1 Summary

The Company Network Design project was successfully executed, achieving key milestones such as a hierarchical model with redundancy, VLAN-based segmentation, inter-VLAN routing, robust security measures, and effective NAT. Testing via Cisco Packet Tracer ensured scalability, security, and efficiency, meeting organizational needs.

5.2 Limitation

1. Relies on Cisco Packet Tracer, which may not fully replicate real-world scenarios.
2. Scalability and heavy load testing are limited in the simulation.
3. Hardware constraints and costs were not fully addressed.
4. Advanced features like IPv6 and cloud integration were not implemented.
5. No real-world security audits were conducted.
6. Energy efficiency was not a focus in this design.

5.3 Future Work

1. Introduce network monitoring tools and advanced security measures.
2. Explore virtualization, IPv6, and advanced routing protocols.
3. Expand wireless networks and integrate with cloud services.
4. Conduct regular security audits and provide ongoing training.
5. Focus on energy-efficient designs.

References

- Cisco Networking Academy: *Routing and Switching Essentials v6 Companion Guide*, Cisco Press, 2016.
- Internal references used throughout the document, like project-specific configurations and methodologies.
- Concepts and tools like Cisco Packet Tracer and other networking protocols (e.g., OSPF, VLANs).